Muhammad Hassan

Drexel Hill, PA 19026 | (484) 474-7700 | msh345@drexel.edu | linkedin.com/muhammad-hassan | Portfolio

EDUCATION

Drexel University | Philadelphia, PA GPA: 3.66
Bachelor of Science in Mechanical Engineering | *Mathematics Minor* Anticipated Graduation: June, 2029
Honors & Awards: NSF AIME Scholar, Drexel Founders Scholar, Fannie M. & James Dewey Hardy Scholarship, Charles Husten Grant, William Utzy Scholarship
Relevant Coursework: CAD (SolidWorks), Mechatronics, Statics, Dynamics, Engr Design, Material Science, Thermodynamics, Calculus I, II, Multivariate Calculus, Differential Equations, Mechanics of Materials, Thermodynamic Analysis, Discrete Mathematics, Fundamentals of Physics I, II, III

PROFESSIONAL EXPERIENCE

HERA | Philadelphia, PA | *Cosmic-Particle Research & HASP Payload Design* **February 2026 – Present**

- Designed and engineered a high-altitude 20kg payload in SolidWorks optimized for structural resistance against multidirectional loading exceeding ≥10 Gs. thermal factors, and vibration profiles.
- Performed structural trade studies comparing tungsten and aluminum support structures to balance high-G mechanical stability, mass constraints, and thermal endurance.
- Co-authored the project technical abstract and presented findings for NASA Pennsylvania Space Grant Consortium documentation.

Exelon | Eddystone, PA | *Engineering Co-op – Construction & Maintenance* **March 2026 – September 2026**

- Developed custom Excel operational tools and work-order tracking dashboards, optimizing data tracking, eliminating workflow redundancies, and generating over $100k+ in estimated annual savings.
- Performed field engineering inspections and observational support alongside utility crews, evaluating power distribution equipment, transformer dimensioning, reclosers, and triple-phase primary lines.
- Cross-functionally supported understaffed operations across separate departments while maintaining original deliverables.

PROJECTS

Infant Medicine Delivery System - *Freshman Design Project*

- Conducted comprehensive analysis and concept generation for a new infant drug delivery system, successfully incorporating a medicine reservoir and dispensing mechanism within a standard pacifier.
- Performed prototyping and fabrication, physically assembling multiple working models to test safety, fluid dynamics, and ergonomic constraints.

Drone Engineering & FAA Protocol Initiative - *Current Team Lead*

- Designing and fabricating a custom mechanical drop-hook attachment mechanism in SolidWorks to recover inaccessible payloads.
- Executing rigorous field flight testing protocols to evaluate flight stability, payload balance, and battery draw under varying load conditions.

For all of my previous projects and current work, please check out my portfolio at muhasofficial.github.io!

TECHNICAL SKILLS

- **CAD & Simulation:** SolidWorks (3D Modeling, Assembly Mating, Exploded Views, Technical Drawings), AutoCAD, FEA Stress/Vibration Analysis, FMEA
- **Data Analysis & Systems Engineering:** Applied Engineering Computations, Dynamic Systems Modeling, Operational Workflow Optimization, Master Data Tracking, Statistical Analysis
- **Mechatronics:** Microcontroller Integration (Raspberry Pi Pico, Arduino) , Sensor & Actuator Interfacing, Motor Drivers (L298N, MOSFETs), Circuit Prototyping
- **Fabrication & Lab Work:** Prototyping, 3D Printing, Benchtop Electronics Testing, Hand & Power Tools, Quality Assurance / SOP Enforcement, Soldering